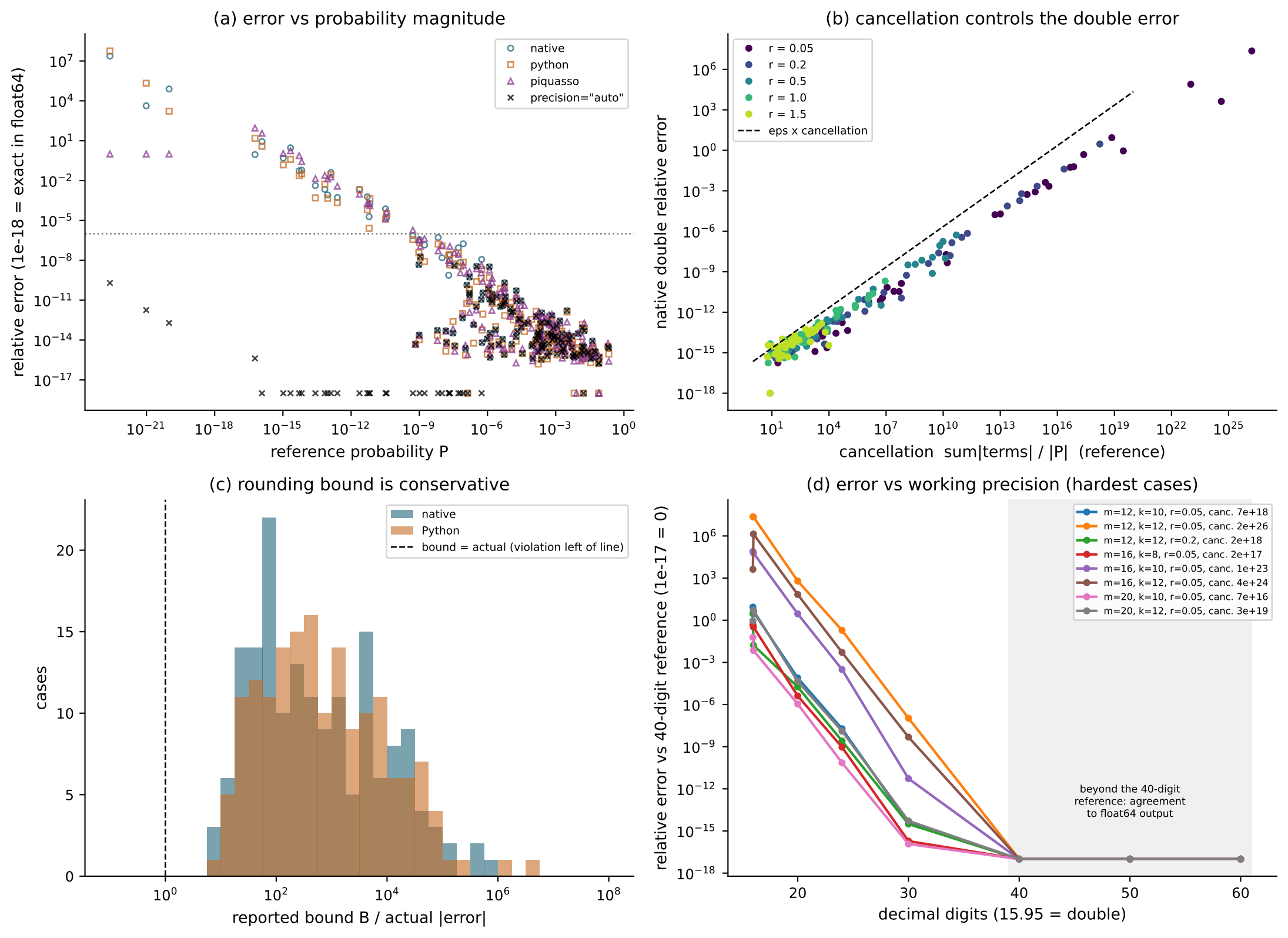
(a) error vs probability magnitude
native
python
piquasso
precision="auto"
relative error (1e-18 = exact in float64)
reference probability P
(b) cancellation controls the double error
r = 0.05
r = 0.2
r = 0.5
r = 1.0
r = 1.5
eps x cancellation
native double relative error
cancellation sum|terms| / |P| (reference)
(c) rounding bound is conservative
native
Python
bound = actual (violation left of line)
cases
reported bound B / actual |error|
(d) error vs working precision (hardest cases)
m=12, k=10, r=0.05, canc. 7e+18
m=12, k=12, r=0.05, canc. 2e+26
m=12, k=12, r=0.2, canc. 2e+18
m=16, k=8, r=0.05, canc. 2e+17
m=16, k=10, r=0.05, canc. 1e+23
m=16, k=12, r=0.05, canc. 4e+24
m=20, k=10, r=0.05, canc. 7e+16
m=20, k=12, r=0.05, canc. 3e+19
beyond the 40-digit reference: agreement to float64 output
relative error vs 40-digit reference (1e-17 = 0)
decimal digits (15.95 = double)